

Hamming code with solved problems

Hamming code is used to detect and correct the error in the transmitted data. So, it is an [error detection and correction code](#). It was originally invented by *Richard W. Hamming* in the year 1950. Hamming codes detect 1-bit and 2-bit errors.

While transmitting the message, it is encoded with the redundant bits. The redundant bits are the extra bits that are placed at certain locations of the data bits to detect the error. At the receiver end, the code is decoded to detect errors and the original message is received.

So before transmitting, the sender has to encode the message with the redundant bits. It involves three steps, as described below.

Encoding the message with hamming code

Selecting the number of redundant bits

The hamming code uses the number of redundant bits depending on the number of information bits in the message.

Let n be the number of information or data bits, then the number of redundant bits P is determined from the following formula,

$$2^P \geq n + P + 1$$

For example, if 4-bit information is to be transmitted, then $n=4$. The number of redundant bits is determined by the trial and error method.

Let $P=2$, we get,

$$2^2 \geq 4 + 2 + 1$$

The above equation implies 4 not greater than or equal to 7. So let's choose another value of $P=3$.

$$2^3 \geq 4 + 3 + 1$$

Now, the equation satisfies the condition. So number of redundant bits, $P=3$. In this way, the number of redundant bits is selected for the number of information bits to be transmitted.

Choosing the location of redundant bits

For the above example, the number of data bits $n=4$, and the number of redundant bits $P=3$. So the message consists of 7 bits in total that are to be coded. Let the rightmost bit be designated as bit 1, the next successive bit as bit 2 and so on.

The seven bits are bit 7, bit 6, bit 5, bit 4, bit 3, bit 2, bit 1.

In this, the redundant bits are placed at the positions that are numbered corresponding to the power of 2, i.e., 1, 2, 4, 8,... Thus the locations of data bit and redundant bit are $D_4, D_3, D_2, P_3, D_1, P_2, P_1$.

Assigning the values to redundant bits

Now it is time to assign bit value to the redundant bits in the formed hamming code group. The assigned bits are called a parity bit.

Each parity bit will check certain other bits in the total code group. It is one with the bit location table, as shown below.

Bit Location	7	6	5	4	3	2	1
Bit designation	D ₄	D ₃	D ₂	P₃	D ₁	P₂	P₁
Binary representation	111	110	101	100	011	010	001
Information / Data bits	D ₄	D ₃	D ₂		D ₁		
Parity bits				P₃		P₂	P₁

Parity bit P_1 covers all data bits in positions whose binary representation has 1 in the least significant position(001, 011, 101, 111, etc.). Thus P_1 checks the bit in locations 1, 3, 5, 7, 9, 11, etc..

Parity bit P₂ covers all data bits in positions whose binary representation has 1 in the second least significant position(010, 011, 110, 111, etc.). Thus P₂ checks the bit in locations 2, 3, 6, 7, etc.

Parity bit P₃ covers all data bits in positions whose binary representation has 1 in the third least significant position(100, 101, 110, 111, etc.). Thus P₃ checks the bit in locations 4, 5, 6, 7, etc.

Each parity bit checks the corresponding bit locations and assign the bit value as 1 or 0, so as to make the number of 1s as even for even parity and odd for odd parity.

Example problem 1

Encode a binary word 11001 into the even parity hamming code.

Given, number of data bits, $n = 5$.

To find the number of redundant bits,

Let us try $P=4$.

The equation is satisfied and so 4 redundant bits are selected.

So, total code bit = $n+P = 9$

The redundant bits are placed at bit positions 1, 2, 4 and 8.

Construct the bit location table.

Bit Location	9	8	7	6	5	4	3	2	1
Bit designation	D ₅	P₄	D ₄	D ₃	D ₂	P₃	D ₁	P₂	P₁
Binary representation	1001	1000	0111	0110	0101	0100	0011	0010	0001
Information bits	1		1	0	0		1		
Parity bits		1				1		0	1

To determine the parity bits

For P1: Bit locations 3, 5, 7 and 9 have three 1s. To have even parity, P1 must be 1.

For P2: Bit locations 3, 6, 7 have two 1s. To have even parity, P2 must be 0.

For P3: Bit locations 5, 6, 7 have one 1s. To have even parity, P3 must be 1.

For P4: Bit locations 8, 9 have one 1s. To have even parity, P2 must be 1.

Thus the encoded 9-bit hamming code is 111001101.

How to detect and correct the error in the hamming code?

After receiving the encoded message, each parity bit along with its corresponding group of bits are checked for proper parity. While checking, the correct result of individual parity is marked as 0 and the wrong result is marked as 1.

After checking all the parity bits, a binary word is formed taking the result bits for P1 as LSB. So formed binary word gives the bit location, where there is an error.

If the formed binary word has 0 bits, then there is no error in the message.

Example problem 2

Let us assume the even parity hamming code from the above example (111001101) is transmitted and the received code is (110001101). Now from the received code, let us detect and correct the error.

To detect the error, let us construct the bit location table.

Bit Location	9	8	7	6	5	4	3	2	1
Bit designation	D5	P4	D4	D3	D2	P3	D1	P2	P1
Binary representation	1001	1000	0111	0110	0101	0100	0011	0010	0001
Received code	1	1	0	0	0	1	1	0	1

Checking the parity bits

For P1 : Check the locations 1, 3, 5, 7, 9. There is three 1s in this group, which is wrong for even parity. Hence the bit value for P1 is 1.

For P2 : Check the locations 2, 3, 6, 7. There is one 1 in this group, which is wrong for even parity. Hence the bit value for P2 is 1.

For P3 : Check the locations 3, 5, 6, 7. There is one 1 in this group, which is wrong for even parity. Hence the bit value for P3 is 1.

For P4 : Check the locations 8, 9. There are two 1s in this group, which is correct for even parity. Hence the bit value for P4 is 0.

The resultant binary word is 0111. It corresponds to the bit location 7 in the above table. The error is detected in the data bit D4. The error is 0 and it should be changed to 1. **Thus the corrected code is 111001101.**